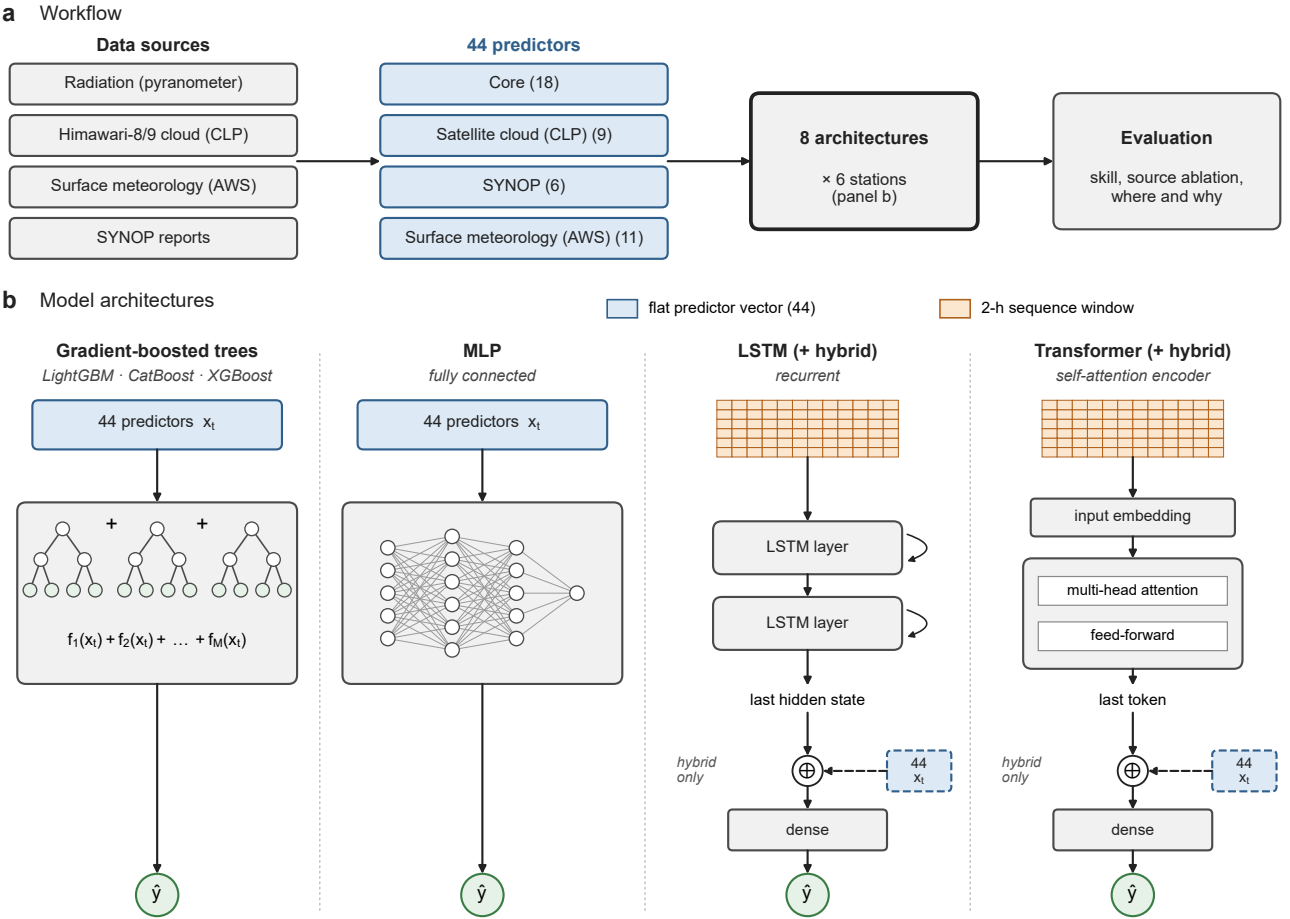

a
Workflow
Data sources
Radiation (pyranometer)
Himawari-8/9 cloud (CLP)
Surface meteorology (AWS)
SYNOP reports
44 predictors
Core (18)
Satellite cloud (CLP) (9)
SYNOP (6)
Surface meteorology (AWS) (11)
8 architectures
× 6 stations
(panel b)
Evaluation
skill, source ablation,
where and why
b
Model architectures
flat predictor vector (44)
2-h sequence window
Gradient-boosted trees
LightGBM · CatBoost · XGBoost
44 predictors x_t
$f_1(x_t) + f_2(x_t) + \ldots + f_M(x_t)$
$\hat{y}$
MLP
fully connected
44 predictors x_t
$\hat{y}$
LSTM (+ hybrid)
recurrent
LSTM layer
LSTM layer
last hidden state
hybrid only
44 x_t
dense
$\hat{y}$
Transformer (+ hybrid)
self-attention encoder
input embedding
multi-head attention
feed-forward
last token
hybrid only
44 x_t
dense
$\hat{y}$